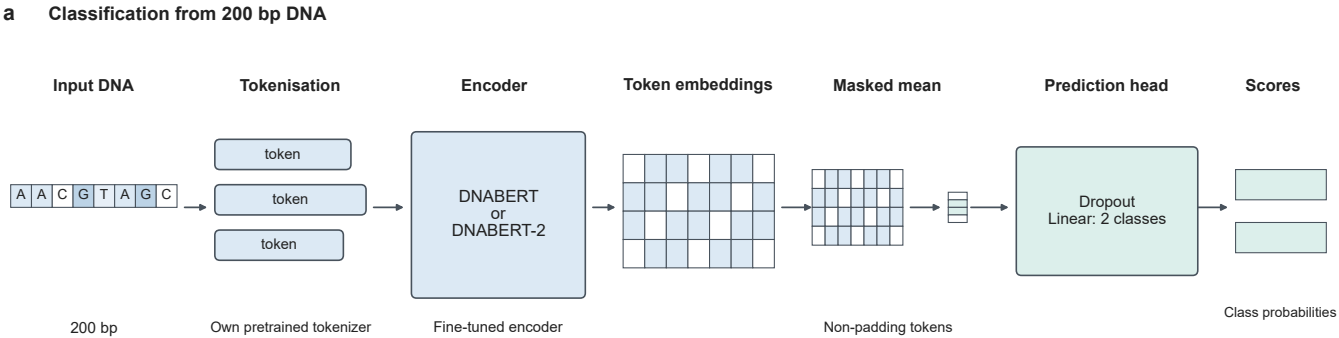
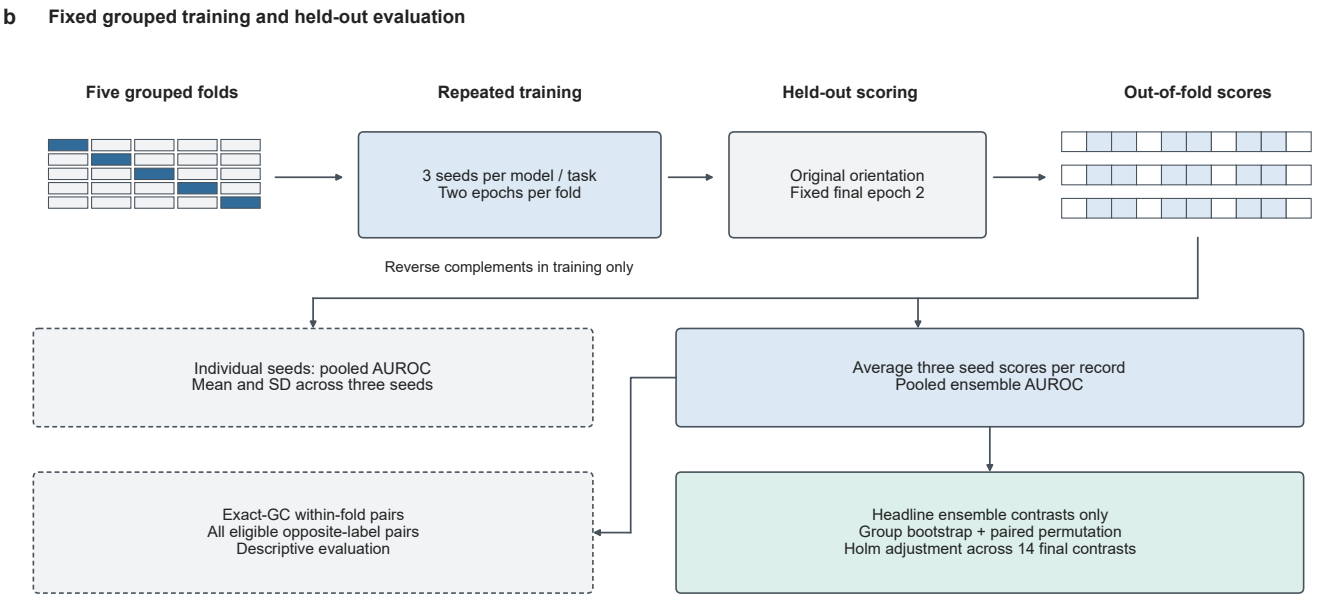

a
Classification from 200 bp DNA
Input DNA
Tokenisation
Encoder
Token embeddings
Masked mean
Prediction head
Scores
A A C G T A G C
token
token
token
DNABERT
or
DNABERT-2
Dropout
Linear: 2 classes
200 bp
Own pretrained tokenizer
Fine-tuned encoder
Non-padding tokens
Class probabilities


b
Fixed grouped training and held-out evaluation
Five grouped folds
Repeated training
Held-out scoring
Out-of-fold scores
3 seeds per model / task
Two epochs per fold
Original orientation
Fixed final epoch 2
Reverse complements in training only
Individual seeds: pooled AUROC
Mean and SD across three seeds
Average three seed scores per record
Pooled ensemble AUROC
Exact-GC within-fold pairs
All eligible opposite-label pairs
Descriptive evaluation
Headline ensemble contrasts only
Group bootstrap + paired permutation
Holm adjustment across 14 final contrasts